

F546 — items 1 & 2 second pass: (1) mixing-depth FIRST-PASS handed to Grace (forward, model-flagged) — from the pinned ladder (F544), the climbing-mode radial expectation $\langle N \rangle_k = E_0/(E_0+k)$ with $E_0=\Delta_{Di}=2$ gives depths $\{N_e, N_\mu, N_\tau\} = \{1, 2/3, 1/2\}$ (electron $N=1$ at the origin $\square$ as a sanity check), a disk-analog (rank-1) first-pass with the rank-2 FK radial correction flagged as the joint next piece — hand to Grace to RUN her CP/mixing machinery, refine after, NOT banked; (2) the up premise ‘depth $\propto \sqrt{(\text{Peirce block dim})}$ ’ (F545) HELD as a strong-lead — a bounded search finds NO clean second observable it forces (none of the block-dim $\sqrt{\cdot}$ -ratios $\sqrt{(N_c/\text{rank})}$, $\sqrt{(n_C/\text{rank})}$, $\sqrt{(n_C/N_c)}$, $\sqrt{(\text{rank}/n_C)}$... lands on an independent known observable at $<1\%$ without fishing), so per Keeper’s item-2 alternative I hold the up honestly (predicts the up-1 depth, forces no second) rather than dress the single-target fit. ITEM 1 (the mixing-sector finish-line unblock): the localization depth = the mean generic-norm $\langle N \rangle$ in the climbing-mode state; for the holomorphic discrete series with lowest weight E_0 at climbing level k , the rank-1 (disk) radial expectation is $\langle N \rangle_k = E_0/(E_0+k)$; with the SOURCED $E_0 = \Delta_{Di} = 2$ (F544): electron $\langle N \rangle_0 = 1$ (origin, sanity $\square$), muon $\langle N \rangle_1 = 2/3 = 0.667$, tau $\langle N \rangle_2 = 1/2 = 0.500$; overlaps $N^{\{n_C/2\}} = 1, 0.363, 0.177$. FLAGGED: this is the disk-analog (rank-1); the exact rank-2 type-IV FK radial correction is the flagged joint next piece — hand these to Grace as a forward FIRST-PASS to run her F498-respecting CP/mixing machinery (Jarlskog, V_{us} , V_{ub} , θ_{23} octant, θ_{12}), then refine with the FK rank-2 model. Differs from the corpus $N_\mu \approx 0.55$ (which was a

$V_{us}=1/\sqrt{20}$ BACK-CALC, target-aware) — my 0.667 is FORWARD (from the sourced E_0), the gap is the rank-2 correction + the fact 0.55 was fitted. NOT banked (model-flagged first-pass). ITEM 2 (ground or hold the up premise): bounded search — Peirce block dims are diagonal=rank=2, off-diagonal $V_{12}=N_c=3$, full $V_5=n_C=5$; the $\sqrt{\cdot}$ -ratios of these $\{\sqrt{(3/2)}=\text{up-1}, \sqrt{(5/2)}, \sqrt{(5/3)}, \sqrt{(2/3)}, \sqrt{(2/5)}, \sqrt{(3/5)}\}$ — only $\sqrt{(N_c/\text{rank})}=\sqrt{(3/2)}$ matches an observable (the up-1, F545, the ORIGINAL target, not independent); none of the others lands on an independent known observable at $<1\%$ without fishing. VERDICT (Keeper's item-2 alternative, taken): HOLD the up as a STRONG-LEAD — it predicts the up-1 depth ($m_u/m_d=\sqrt{(3/14)}$, 0.1%, from the Peirce block-size reflection, F545) but forces NO clean second observable, so I do NOT dress the single-target fit; the load-bearing premise 'depth $\propto \sqrt{(\text{Peirce block dim})}$ ' stays a named-but-unforced premise. This is a precise clean assessment ('predicts one, forces no second'), a legitimate advance per the mandate. For Casey (two updates: first, the mixing sector's finish line — I handed Grace a forward first estimate of the three lepton 'depths' $\{1, 2/3, 1/2\}$ computed from the now-pinned ladder (the electron comes out exactly at the origin, depth 1, which is the right sanity check), so her CP and mixing machinery can RUN now; it's a first pass from the simplest (disk) version of the radial math, with the exact rank-2 refinement flagged as the next joint step — not banked, but enough to unblock her. Second, on the light up-quark: I checked honestly whether the 'depth scales as $\sqrt{(\text{block size})}$ ' idea I used for it predicts anything ELSE we can test, and it doesn't — it nails the up but forces no second number, so per the discipline

I'm holding it as a strong lead rather than dressing up a one-target fit. A clean 'predicts one, forces no second' is the honest verdict), Grace (HANDOFF — forward first-pass depths $\{N_e, N_\mu, N_\tau\} = \{1, 2/3, 1/2\}$ from the pinned ladder ($E_0 = \Delta_{Di} = 2$, F544), disk-analog rank-1, rank-2 FK correction flagged; RUN your CP/mixing machinery on these — Jarlskog, V_{us} , V_{ub} , θ_{23} octant, θ_{12} — and report forward-or-can-fail; NOT banked, refine with the rank-2 model after; note these differ from the old $N_\mu \approx 0.55$ which was a V_{us} back-calc, mine are forward), Keeper (F546: item 1 — forward first-pass depths $\{1, 2/3, 1/2\}$ to Grace (disk-analog, rank-2 correction flagged, not banked); item 2 — up premise HELD as strong-lead per your alternative, bounded search finds no clean second observable it forces, 'predicts one forces no second', not dressing the fit), Elie (the depth first-pass $\{1, 2/3, 1/2\}$ feeds the mixing; separately your three $\sqrt{}$ -flavors + my block-dim $\sqrt{(N_c/\text{rank})}$ — I checked and the block-dim $\sqrt{}$ forces only the up-1, no second, held as strong-lead), Cal (F546: item 1 mixing-depth first-pass $\{1, 2/3, 1/2\}$ to Grace (model-flagged, not banked); item 2 up premise held strong-lead (no clean second observable, not fitting); both target-innocent-where-forward, honestly tiered).

Lyra (Claude Opus 4.8)

2026-07-15 Wednesday (date-verified)

F546 — Items 1 & 2 second pass: mixing-depth handoff to Grace, and holding the up premise honestly

Item 1 — the mixing-sector unblock: forward first-pass depths to Grace

The localization depth = the mean generic-norm $\langle N \rangle$ in the climbing-mode state. For the holomorphic discrete series with lowest weight E_0 at climbing level k , the rank-1 (disk) radial expectation is $\langle N \rangle_k = E_0 / (E_0 + k)$. With the sourced $E_0 = \Delta_{Di} = 2$ (F544):

gen	$\langle N \rangle_k = E_0/(E_0+k)$	overlap $N^{n_C/2}$
electron	$2/2 = 1$ (origin $\boxed{?}$)	1
muon	$2/3 = 0.667$	0.363
tau	$2/4 = 0.500$	0.177

Flagged: this is the disk-analog (rank-1); the exact rank-2 type-IV FK radial correction is the joint next piece. Handed to Grace as a forward first-pass to run her F498-respecting CP/mixing machinery (Jarlskog, V_{us} , V_{ub} , θ_{23} octant, θ_{12}), then refine. The electron landing exactly at the origin (depth 1) is the sanity check. Differs from the corpus $N_\mu \approx 0.55$ (which was a $V_{us}=1/\sqrt{20}$ *back-calc*, target-aware) — mine is forward from the sourced E_0 ; the gap is the rank-2 correction + the fact 0.55 was fitted. Not banked (model-flagged first-pass).

Item 2 — the up premise: held honestly as a strong-lead

Bounded search: does “depth $\propto \sqrt{(\text{Peirce block dim})}$ ” (F545) force any independent observable beyond the up-1? Peirce block dims: diagonal = rank = 2, off-diagonal $V_{12} = N_c = 3$, full $V_5 = n_C = 5$. The $\sqrt{\cdot}$ -ratios $\{\sqrt{(3/2)}, \sqrt{(5/2)}, \sqrt{(5/3)}, \sqrt{(2/3)}, \sqrt{(2/5)}, \sqrt{(3/5)}\}$ — only $\sqrt{(N_c/\text{rank})} = \sqrt{(3/2)}$ matches an observable (the up-1, the *original* target, not independent); none of the others lands on an independent known observable at $<1\%$ without fishing.

Verdict (Keeper’s item-2 alternative, taken): hold the up as a strong-lead — it predicts the up-1 depth ($m_u/m_d = \sqrt{(3/14)}$, 0.1%, from the Peirce block-size reflection, F545) but forces no clean second observable, so I do not dress the single-target fit. A precise “predicts one, forces no second” is the honest verdict — a legitimate advance.

Plain-language version

Two updates. First, the mixing sector’s finish line: I handed Grace a forward first estimate of the three lepton “depths” $\{1, 2/3, 1/2\}$ from the now-pinned ladder (the electron comes out exactly at the origin, depth 1 — the right sanity check), so her CP and mixing machinery can run now. It’s a first pass from the simplest version of the radial math, with the exact refinement flagged as the next joint step — not banked, but enough to unblock her. Second, on the light up-quark: I checked honestly whether the “depth scales as $\sqrt{(\text{block size})}$ ” idea predicts anything *else* we can test, and it doesn’t — it nails the up but forces no second number. So per the discipline, I’m holding it as a strong lead rather than dressing up a one-target fit.

— Lyra, 2026-07-15. F546: item 1 — forward first-pass depths $\{1, 2/3, 1/2\}$ to Grace ($\langle N \rangle_k = E_0/(E_0+k)$, $E_0 = \Delta_{Di}=2$, F544; electron=1 origin $\boxed{?}$; disk-analog, rank-2 correction flagged, not banked). Item 2 — up premise held as strong-lead (bounded search: no clean second observable it forces, don’t dress the fit). Both honestly tiered.